# MILITARY SPECIFICATION
# COOLER, LIQUID, ELECTRONIC EQUIPMENT
# HD-957( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

## 1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Electronic Equipment Liquid Cooler (LC) HD-957 ( )/AWM-23(V).

1.2 Classification. The LC shall consist of a cabinet assembly and a remote switching control. The cabinet assembly shall contain a coolant circulation loop, a chilled water loop, local controls and indicators, and electronic subassemblies.

a. Cabinet assembly, major elements

| Nomenclature | Reference Paragraph |
|---|---|
| Pumping Unit - Hydraulic | 3.6.1 |
| Chiller | 3.6.2 |
| Control Panel, External | 3,6.3 |
| Gageboard, Internal | 3.6.4 |

b. Remote switching control

| Nomenclature | |
|---|---|
| Control, Remote Switching | 3,6.5 |

## 2. APPLICABLE DOCUMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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2.1 General. The following documents of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) is the applicable general specification for the LC described herein (see 6.6).

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-M-7793 | Meter, Time totalizing |
| MIL-R-16743 | Refrigerating Plants and Systems, Mechanical and Refrigerating System Components - Dichlorodifluoromethane Type 12 - Naval Shipboard |
| MIL-T-81734(AS) | Test Set, Airborne Weapon Control System, AN/AWM-23(V) |

Hughes Aircraft Company

| | |
|---|---|
| HMS 20-1458 | Material Specification, Fluid, Dielectric Coolant, Silicate, Ester Base |
| HP 10-27 | Hughes Process, Dielectric Coolant and Hydraulic Fluid and Operating Systems, Requirements for |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-761 | Electric Power, Alternating Current for Shipboard Use, Characteristics and Utilization of |

PUBLICATION

Hughes Aircraft Company

| | |
|---|---|
| 3282244 | Acceptance Test Procedure, Electronic Equipment Liquid Cooler |

2.2 Availability of documents. Applicable (see 6.6)

2.3 Precedence of documents, When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any other document
b. This Specification - This specification shall have precedence over all applicable subsidiary documents (see 6.6)
c. Referenced documents - Any document referenced in this specification shall have precedence over all applicable documents referenced therein.



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3. REQUIREMENTS

3.1 Materials, parts, and processes. Not applicable. Unless otherwise specified herein, materials, parts and processes shall be in accordance with MIL-R-16743.

3.2 Design and construction. Applicable.

3.2.1 Design features. Not applicable. Unless otherwise specified herein, design features shall be in accordance with MIL-R-16743.

3.2.1.1 Method of operation, Applicable.

3.2.1.2 Commercial test equipment. Applicable.

3.2. 1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability, Applicable,

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

3.2.1.8 Forced-liquid cooling. The LC shall supply cooling fluid to the test stations and weapon replaceable assemblies (WRAs) listed in Table I at the required operating temperature, pressure, and listed flow rate.

3.2.1.9 Space and weight limitations. The cabinet assembly shall not exceed 36 inches by 30 inches by 72 inches. The remote control panel shall not exceed 8 inches by 4 inches by 8 inches. The weight shall not exceed 1500 pounds.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary electrical power. Not applicable.

3.2.1.12 Interface. The LC shall provide the interfaces specified below.

3.2. 1.12.1 Test station interfaces. The LC shall interface with Test Station, Radio Frequency OJ-213( )/AWM-23(V), Test Station, Low Frequency OJ-215( )/AWM-23(V) and OJ-379( )/AWM-23(V) and Stand, Fill-Drain, Coolant System MT-4730( )/AWM-23(V) through two utility coolant lines (supply and return).

3.2. 1.12.2 Potable water interface. The LC shall interface with the potable water through two utility water lines (supply and return).

3.2.1. 12.3 Electrical power interface. The LC shall interface with the electrical power specified in 3.2.2.1 through utility power lines.



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TABLE I

AN/AWG-9 WRAs AND AN/AWM-23
ELEMENTS SERVICED BY THE LIQUID COOLER

| ITEM | | MAXIMUM * FLOW RATE GPM |
|---|---|---|
| TEST STATION, RADIO FREQUENCY | OJ-213( )/AWM-23(V) | 6.8 |
| a. Oscillator, Radio Frequency | O-1633( )/AWG-9 | 0.2 |
| b. Transmitter, Radar | T-1224( )/AWG-9 | 4.8 |
| c. Power Supply | PP-6770( )/AWG-9 | 0.8 |
| d. Power Supply | PP-6771( )/AWG-9 | 0.8 |
| e. Receiver, Radar | R-1785( )/AWG-9 | 0.3 |
| f. Dummy Load, Electrical | DA-630( )/AWM-23(V) | 5.0 |
| TEST STATION, LOW FREQUENCY | OJ-215( )/AWM-23(V) | 2.5 |
| and | OJ-379( )/AWM-23(V) | 2.5 |
| a. Antenna | AS-2764( )/AWG-9 | 1.0 |
| Antenna Servo Electronic Subassembly (when tested separately) | | 1.0 |
| b. Receiver, Infrared | R-1786( )/AWG-9 *** | 0.5 |
| c. Processor, Data | CV-2947( )/AWG-9 *** | 0.1 |
| d. Dummy Load, Electrical | DA-631( )/AWM-23(V) | 1.3 ** |
| e. Dummy Load, Electrical | DA-632( )/AWM-23(V) | |
| STAND, FILL-DRAIN COOLANT | MT-4730( )/AWM-23(V) | 3.0 |

* Combined flow rates for test station dummy loads and WRAs under test will not exceed the maximum flow rate listed for the test station.

** DA-631 and DA-632 are connected in series.
*** Tested on OJ-215( )/AWM-23(V) only.



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3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Applicable. The LC shall operate from the utility electrical power forms listed in Table II.

TABLE II

LC ELECTRICAL POWER

| Item | Input Power Forms | Maximum Load |
|---|---|---|
| 1 | +29 ±4 Vdc | 10 amperes |
| 2 | 440 Vac 60 Hertz (Hx), 3-phase, 3-wire, ungrounded | 11.0 kilovolt-amperes (kVA), 0.8 to 1.0 lagging power factor (pf) |
| 3 | 220 Vac, 60 Hz, 3-phase, 3-wire, ungrounded, per MIL-STD-761 Type I. (Alternative power form for van application.) | 11 kVA, 0.8 to 1.0 lagging pf |

3.2.2.2 Clean air. Not applicable.

3.2.2.3 Potable water. Applicable.

3.2.2.4 Liquid coolant. Applicable.

3.2.2.5 Helium. Not applicable.

3.2.2.6 Nitrogen. Applicable.

3.2.2.7 Hydraulic fluid. Not applicable.

3.2.3 Service conditions.

3.2.3.1 Electrical. Applicable.

3.2.3.2 Environmental. Applicable. In addition, the LC shall withstand surface ships motion of ±15 degrees roll, average period 16 seconds and ±4 degrees pitch, average period 4 seconds. There-shall be no restrictions on the installation of the LC regarding orientation with respect to roll and pitch axes of the ship.

3.2.3.2.1 Organizational level. Not applicable.



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3.2.3.2.2 Intermediate level. Applicable.

3.2.4 Maintainability. Applicable.

3.2.4.1 Mean time to repair. Applicable.

3.2.4.2 Test points. Appl&ab&

3.2.4.3 Shop replaceable assemblies. Applicable.

3.2.4.4 Self test capability. Not applicable.

3.2.4.5 Access. All maintenance and calibration operations shall be via access to the front of the liquid cooler within a 36 inch aisle space or to the top of the liquid cooler at the front within the space below a 76 inch overhead. The liquid cooler shall provide for access from the front for removal and replacement of, but not restricted to, the following items without requirements to solder, weld, or saw:

a. Temperature gages, pressure gages, filter elements, and strainer elements
b. Fuses or circuit breakers
Valves and piping subassemblies
c. Low and high pressure control switches
e. Pumping unit
f. Chiller

3.2.4.6 Internal flow bypass. Provisions shall be incorporated for operation of the pumping loop Independent of the external LC load. An orifice which produces a pressure drop of 75 psi at 13 gpm shall be provided in series with a metering valve for flow adjustment across the output coolant lines. Flow monitoring capability shall be provided to permit pressure regulator adjustment.

3.2.4.7 Self-maintainability features. The controls of 3.6.3 and 3.6.4 shall be provided to facilitate installation and maintenance of the LC.

3.2.5 Reliability. The mean time between failures shall be 500 hours minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagnetic interference (EMI). Not applicable.

.3.2.7 Utilities protection. Applicable.

3.2.7.1 Liquid cooling protection. Applicable. Liquid cooling protection shall be provided by automatic shut down of the LC.



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3.2.7.2 Clean air protection. Not applicable.

3.2.7.3 Electrical power protection, Applicable.

3.2.7.3.1 Emergency shutdown switch. An emergency shutdown switch shall be provided for use by the operator. When the emergency shutdown switch is activated the load side of all LC primary circuit breakers shall be disconnected from their respective loads. The switch utilized for this application shall be of the pushbutton activated, momentary type with the pushbutton finished in red. The front panel area around the pushbutton shall be marked with alternating orange-yellow and black stripes. The switch shall be centrally located on the external control panel of the LC in an area which is easily accessible to the operator during normal operation.

3.2,8 Calibration cycle. Applicable.

3.2.9 Power switching and coolant control. The LC shall provide the necessary switching and hydraulic devices to control utility power forms, liquid coolant and potable water.

3.2.10 Displays. Displays provided on the cabinet assembly and on the remote switching control shall provide an indication of LC operational status.

3.2.11 Installation. The LC shall be capable of supplying coolant to Test Station, Radio Frequency OJ-213( )/AWM-23(V) Test Station, Low Frequency OJ-215( )/AWM-23(V) and OJ-379( )/AWM-23(V) and Stand, Fill-Drain, Coolant System MT-4730( )/AWM-23(V) as follows:

a. A maximum of 30 psi drop in the supply and return lines as a closed loop.
b. No vertical head difference between the LC and the test stations.

3.3 Performance.

3.3.1 General. The LC shall provide liquid cooling with the following parameters:

a. Supply pressure. Pressure of the coolant at the output of the LC shall be ajustable between 150 and 95 psig and shall be accurate to $^{+10}_{-10}$ psig of the setting. Pressure shall be maintained within $^{+5}_{-5}$ psi for any flow between 1 and 13 gpm output of the unit.
b. Total heat load. The LC shall have the capacity to remove a maximum heat load of 80,000 British thermal units per hour (Btu/hour) at a coolant flow of 8 gpm, and 13,000 Btu/hour at a coolant flow of 1.3 gpm.
c. Coolant temperature. Input to LC SO to 125°F
Output of LC 65 $^{+20°F}_{-15°F}$
d. Particulate filtration. 10 microns absolute, full flow
e. Reconditioning filtration. See 3.4.3



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3,3.2 Organizational level functional requirements. Not applicable.

3.3.3 Intermediate level functional requirements. Applicable.

3.3.4 Warm-up time. Applicable.

3.4 Detailed functional requirements.

3.4.1 Coolant circulation loop. Reservoir-stored coolant shall be filtered and then pumped to the chiller for heat exchange with the chilled water. Temperature and pressure gages shall provide for monitoring of output coolant pressures and temperature. A bleed valve shall be provided at the high point and a drain valve shall be provided at the low point.

3.4.2 Particulate filter. A full-flow, depth type disposable filter shall be located in the coolant supply line. The filter shall have an electrical differential pressure indicator to provide an indication of excessive dirt accumulation and provide automatic shut-off of the LC.

3.4.3 Reconditioning filtration, The LC shall continuously circulate the coolant at a rate of 1 to 3 gpm through a reconditioning filtration loop consisting of the following:

a. Orifice - to limit pressure and flow
b. Refining pack filter - Luber-finer Part No. 3132 or equivalent
c. Desiccant filter - Luber-finer Part No. 4493 or equivalent

3.4.4 Coolant reservoir. The coolant reservoir of the LC shall be capable of the following:

| | | |
|---|---|---|
| a. | Volumetric coolant capacity | 2.7 cubic feet minimum (20 gallons nominal) |
| b. | Nitrogen pressurization | 15 psig nominal from nitrogen cylinder specified in 3,6,7 |
| c. | Safety relief valve setting | 15 psig maximum |
| d. | Proof pressure of reservoir | 20 psig maximum |
| e. | Burst pressure | 40 psig minimum |

3.4.5 Controls and indicators. Controls and indicators shall be provided at the cooling stand. The controls and indicators shall provide monitoring and control of the LC functions.

3.5 Workmanship, Applicable.

3.6 LC elements. The LC shall contain the following elements.

3.6.1 Pumping unit-hydraulic. The pumping unit shall deliver reservoir stored coolant to the chiller and then to the test stations via the utility coolant lines. The pumping unit motor shall operate from 220/440 Vat, 3-phase, 60 Hz power.



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3.6.2 Chiller. The chiller shall act as a heat exchanger between the chilled water and the liquid coolant loop,

3.6.3 Control panel, external. The following controls and indicators shall be provided:

a. Local power on indicator lamp
b. Coolant supply pressure gage 0-300 psi
c. Coolant supply temperature gage, +20 to +180°F
d. Coolant return temperature gage, +20 to +180°F
e. Emergency off switch (see 3.2,7.3.1)
f. Coolant reservoir level sight gage
g. Lamp test switch
h. High coolant supply temperature indicator lamp
i. Low coolant reservoir level indicator lamp
j. High particulate filter pressure drop indicator lamp
k. Potable water supply manual shut-off valve
l. Potable water return manual shut-off valve
m. Coolant supply manual shut-off valve
n. Coolant return manual shut-off valve

3.6.4 Gageboard, internal. The following controls and indicators shall be provided:

a. Coolant flow meter, 0 to 12 gpm
b. Chilled water flow meter, 0 to 40 gpm
c. Chilled water inlet pressure gage, 0 to 300 psi
d. Chilled water inlet temperature gage, +20 to 180°F
e. Chilled water outlet temperature gage, +20 to 180°F
f. Coolant supply pressure regulator, 10 to 150 psi
g. Coolant temperature bulb well regulator
h. Particulate filter differential pressure gage, 0 to 160 psi
i. Refining pack filter differential pressure gage, 0 to 160 psi
j. Desiccant filter differential pressure gage, 0 to 160 psi
k. Internal flow bypass metering ball valve
l. Coolant reservoir nitrogen pressure regulator
m. Nitrogen cylinder pressure gage, 0 to 300 psi
n. Nitrogen reservoir pressure gage, 0 to 30 psi
o. Nitrogen shut-off valve
p. Power on switch/circuit breaker, 30 amperes

3.6.5 Control, remote switching. A remote switching control shall be provided to control and monitor liquid coolant at a remote position from the LC. Interconnection between the remote switching control and the LC shall be made by a 30 $^{+1}_{-0}$ foot electrical cable. The following controls and Indicators shall be located on the panel:

a. Power on switch
b. Power off switch
c. Power on indicator light
d. Malfunction indicator light indicating any out of tolerance condition of coolant supply temperature, particulate filter differential pressure, or coolant reservoir low level.



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3.6.6 Elapsed time meter. An elapsed time meter, conforming to MIL-M-7793, shall be provided. The meter shall be readily visible during operation of the LC and shall be capable of registering 9,999 hours.

3.6.7 Nitrogen tank. The nitrogen tank shall consist of a cylinder and shut off valve.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable,

4.2.1 Classification, Applicable.

4.2.2 Individual items. Applicable.

4.2.2.1 Vendor's facility, Tests performed in accordance with Table III may be accomplished at the vendor's facility. Evidence of satisfactory completion shall consist of acceptance plan data sheets containing evidence of compliance of operation to the expected values.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

4.2.5 Acceptance test plan. Applicable. Table III shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance, Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in test. Applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.



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TABLE III

ACCEPTANCE TEST PLAN FOR COOLER, LIQUID,
ELECTRONIC EQUIPMENT
HD-957( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate the functional operation of the liquid cooler.

2.0 TEST CONDITIONS

a. Test Equipment

(1) Metering Valves - Water and coolant lines.
(2) Heat Exchanger and Heat Source - Capable of exchanging 80,000 BTU per hour minimum to HMS 20-1458 coolant at 8 gpm (method for monitoring exchange rate is also required).
(3) Piping and Fittings (1 inch diameter) - For connecting liquid cooler, flow meter, metering valve, heat exchanger and micron filter.

Note: All piping and fittings must be compatible with HMS 20-1458 coolant and in accordance with HP 10-27.

b. Utilities Required (information only, 4.2.8)

(1) 440V, 60 Hz three phase, three wire power as specified by MIL-C-81952(AS).
(2) 30 gpm cooling water as specified by MIL-T-81734(AS).

3.0 TESTS In accordance with Acceptance Test Procedure 3282244

a. Perform operational run-in tests.
b. Perform pressure and temperature tests in accordance with the acceptance test procedure.

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The LC is intended for intermediate level use to supply cooling fluid to the test stations and WRAs listed in Table I.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.



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6.4 Nomenclature cross-reference. The following is a cross reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

| | Official Nomenclature | Contractor Part Number |
|---|---|---|
| a. | Test Station, Radio Frequency OJ-213( )/AWM-23(V) | 493150-1XX |
| b. | Test Station, Low Frequency OJ-215( )/AWM-23(V) OJ-379( )/AWM-23(V) | 493350-1XX |
| c. | Cooler, Liquid, Electronic Equipment HD-957( )/AWM-23(V) | 493711-1XX |
| d. | Dummy Load, Electrical DA-630( )/AWM-23(V) | 3255628-XXX |
| e. | Dummy Load, Electrical DA-631( )/AWM-23(V) | 3069350-XXX |
| f. | Dummy Load, Electrical DA-632( )/AWM-23(V) | 3069320-XXX |
| g. | Oscillator, Radio Frequency 0-1633( )/AWG-9 | 481001-1XX |
| h. | Transmitter, Radar T-1224( )/AWG-9 | 481011-1XX |
| i. | Power Supply PP-6770( )/AWG-9 | 481013-1XX |
| j. | Power Supply PP-6771( )/AWG-9 | 481014-1XX |
| k. | Receiver, Radar R-1785( )/AWG-9 | 481022-1XA |
| l. | Antenna, AS-2764( )/AWG-9 | 481031-1XX |
| m. | Receiver, Infrared R-1786( )/AWG-9 | 481101-1XX |
| n. | Processor, Data CV-2947( )/AWG-9 | 481120-1XX |
| o. | Stand, Fill-Drain, Coolant System MT-4730( )/AWM-23(V) | 493715-1XX |

6.5 Liquid cooler. Applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-C-81952(AS) are the same as MIL-T-81734(AS) the word “applicable” is used in this specification. When the term “not applicable” is used in this specification, it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met. See 6.6 of MIL-T-81734(AS).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply:

| | |
|---|---|
| BTU | British thermal unit |
| CVA | Carrier aircraft attack |
| GPM | Gallons per minute |
| LC | Liquid cooler |
| PSI | Pounds per square inch |
| PSIG | Pounds per square inch gage |

6.8 Convenience outlet power. Not applicable.

6.9 Changes from previous issue are not marked.

Custodian: Navy-AS Preparing Activity: Navy-AS project No, 4935-N017



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DOCUMENT IDENTIFIER AND TITLE

MIL-C-81952B(AS) Cooler, Liquid, Electronic Equipment HD-957( )/AWM-23(V)

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
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| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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